

Identifying errors in proofs - Set 1

TMUA practice questions by topic. Specification Err1, Err2. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 A student attempts to solve the equation $\sqrt{x+3} = x-3$ for real x .

- I Squaring both sides: $x+3 = (x-3)^2 = x^2 - 6x + 9$.
- II So $x^2 - 7x + 6 = 0$.
- III So $(x-1)(x-6) = 0$.
- IV Hence the solutions are $x = 1$ and $x = 6$.

Which of the following best describes the solution?

- A The solution is wrong, and the first error occurs on line IV.
- B The solution is completely correct.
- C The solution is wrong, and the first error occurs on line I.
- D The solution is wrong, and the first error occurs on line II.
- E The solution is wrong, and the first error occurs on line III.

- 2 Here is a purported proof that $1 = 2$.

- I Let $a = b$, with $a \neq 0$.
- II Then $a^2 = ab$.
- III So $a^2 - b^2 = ab - b^2$, i.e. $(a-b)(a+b) = b(a-b)$.
- IV Dividing both sides by $(a-b)$: $a+b = b$.
- V So $2b = b$, and hence $2 = 1$.

On which line does the first error occur?

- A IV
- B II
- C III
- D V
- E I

3 A student tries to solve $\sin \theta = \sin 40^\circ$ for $0^\circ \leq \theta < 360^\circ$.

I Since $\sin \theta = \sin 40^\circ$, we have $\theta = 40^\circ$.

II The sine function has period 360° , so $\theta = 40^\circ + 360^\circ k$.

III In the given interval the only solution is $\theta = 40^\circ$.

Which of the following is correct?

A The first error is on line I: $\sin \theta = \sin \alpha$ does not imply $\theta = \alpha$; $\theta = 140^\circ$ is also a solution.

B The solution is completely correct.

C The first error is on line II: the period of sine is 180° .

D The first error is on line III: $\theta = 400^\circ$ should be included.

E The first error is on line II: it should be $\theta = 40^\circ + 180^\circ k$.

4 Claim: for all real x , if $x^2 > 9$ then $x > 3$.

I Suppose $x^2 > 9$.

II Taking square roots of both sides, $x > 3$.

III This proves the claim.

Which of the following is true?

A The claim is false; the first error is on line II, since $x = -4$ satisfies $x^2 > 9$ but not $x > 3$.

B The claim and proof are both correct.

C The claim is true but line II should say $|x| > 3$ and then deduce $x > 3$.

D The claim is false; the first error is on line I.

E The claim is true; the first error is on line III.

5 A student proves that the equation $x^2 + 2x + 3 = 0$ has no real solutions.

I Complete the square: $x^2 + 2x + 3 = (x + 1)^2 + 2$.

II $(x + 1)^2 \geq 0$ for all real x .

III So $(x + 1)^2 + 2 \geq 2 > 0$ for all real x .

IV Hence the equation has no real solutions.

Which of the following is correct?

A The proof is completely correct.

B The first error is on line I.

C The first error is on line II: $(x + 1)^2$ can be negative.

D The first error is on line III.

E The first error is on line IV: a positive expression can still equal zero.

6 Claim: if $a > b$ then $\frac{1}{a} < \frac{1}{b}$.

I Suppose $a > b$.

II Dividing both sides by ab : $\frac{1}{b} > \frac{1}{a}$.

III So $\frac{1}{a} < \frac{1}{b}$.

Which of the following is correct?

- A The claim is false in general and the first error is on line II, since ab may be negative or zero.
- B The claim and proof are correct.
- C The first error is on line III.
- D The claim is true but the first error is on line I.
- E The claim is false and the first error is on line III.

7 A student solves $x^2 = 4x$.

I Divide both sides by x : $x = 4$.

II So the only solution is $x = 4$.

Which of the following is correct?

- A The first error is on line I: dividing by x discards the solution $x = 0$.
- B The solution is completely correct.
- C The first error is on line II.
- D The first error is on line I: $x^2 \div x = 2x$.
- E The first error is on line I: it should give $x = 2$.

8 Claim: if x and y are real with $x^2 = y^2$, then $x = y$.

Which of the following is a counterexample to the claim?

- A $x = -1, y = -1$
- B $x = 1, y = 2$
- C $x = 2, y = -2$
- D $x = 0, y = 0$
- E $x = 2, y = 2$

- 9 A student argues: 'Every prime number is odd, because if it were even it would be divisible by 2 and so not prime.' Which of the following is correct?
- A The argument is wrong because 2 is even and prime: being divisible by 2 does not stop 2 itself being prime.
- B The argument is correct.
- C The argument is wrong because some odd numbers are not prime.
- D The argument is wrong because 1 is odd and not prime.
- E The argument is wrong because 0 is even.

- 10 A student solves $2\cos^2 x = \cos x$ for $0 \leq x < 2\pi$.

I Divide both sides by $\cos x$: $2\cos x = 1$.

II So $\cos x = \frac{1}{2}$.

III So $x = \frac{\pi}{3}$ or $x = \frac{5\pi}{3}$.

How many solutions has the student missed, and where is the first error?

- A Two solutions missed; the first error is on line I.
- B No solutions missed; the working is correct.
- C One solution missed; the first error is on line III.
- D Two solutions missed; the first error is on line III.
- E One solution missed; the first error is on line I.

- 11 A student solves $x^2 - 5x + 6 > 0$ as follows.

I $x^2 - 5x + 6 = (x - 2)(x - 3)$.

II So $(x - 2)(x - 3) > 0$.

III So $x - 2 > 0$ and $x - 3 > 0$.

IV So $x > 3$.

Which of the following is correct?

- A The first error is on line III: a product is positive when both factors are positive **or** both are negative, so $x < 2$ is missed.
- B The solution is completely correct.
- C The first error is on line I.
- D The first error is on line IV: it should be $x > 2$.
- E The first error is on line II.

- 12 A student solves the inequality $\frac{x+1}{x-2} < 3$.
- I** Multiply both sides by $(x-2)$: $x+1 < 3x-6$.
- II** So $7 < 2x$.
- III** So $x > \frac{7}{2}$.

Which of the following is correct?

- A** The first error is on line I: multiplying by $x-2$ is only valid as written when $x-2 > 0$.
- B** The solution is completely correct.
- C** The first error is on line II: it should be $7 < 4x$.
- D** The first error is on line III: it should be $x < \frac{7}{2}$.
- E** The first error is on line I: you cannot multiply an inequality by an expression.